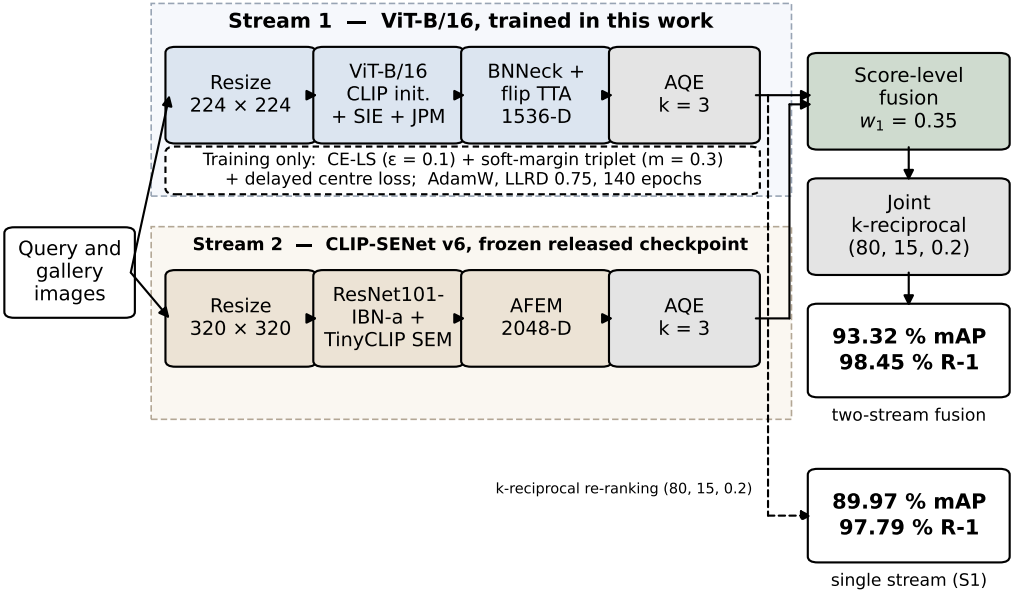
Stream 1 — ViT-B/16, trained in this work
Resize 224 × 224
ViT-B/16 CLIP init. + SIE + JPM
BNNeck + flip TTA 1536-D
AQE k = 3
Training only: CE-LS (ε = 0.1) + soft-margin triplet (m = 0.3) + delayed centre loss; AdamW, LLRD 0.75, 140 epochs
Query and gallery images
Stream 2 — CLIP-SENet v6, frozen released checkpoint
Resize 320 × 320
ResNet101-IBN-a + TinyCLIP SEM
AFEM 2048-D
AQE k = 3
Score-level fusion $w_1 = 0.35$
Joint k-reciprocal (80, 15, 0.2)
93.32 % mAP
98.45 % R-1
two-stream fusion
k-reciprocal re-ranking (80, 15, 0.2)
89.97 % mAP
97.79 % R-1
single stream (S1)